Theorem 1.4: The product of an odd function and an even function is an odd function.

Proof. Let f be an odd function and g be an even function. Let $h = fg$. Now
$h(-x) = (fg)(-x) = f(-x)g(-x) = -f(x)g(x)$ (as f is odd and g is even)
$= -h(x)$
Thus h is an odd function. This shows that fg is an odd function. ■



If one function is not odd then don't think that the function is an even function. There are plenty of functions which are neither even nor odd.



Exercise - 1.3

1. Suppose that 120 students are studying in 4 sections of eleventh standard in a school. Let A denote the set of students and B denote the set of the sections. Define a relation from A to B as "x *related to* y *if the student* x *belongs to the section* y". Is this relation a function? What can you say about the inverse relation? Explain your answer.
2. Write the values of f at $-4, 1, -2, 7, 0$ if

$$f(x) = \begin{cases} -x+4 & \text{if } -\infty < x \leq -3 \\ x+4 & \text{if } -3 < x < -2 \\ x^2 - x & \text{if } -2 \leq x < 1 \\ x - x^2 & \text{if } 1 \leq x < 7 \\ 0 & \text{otherwise} \end{cases}$$

3. Write the values of f at $-3, 5, 2, -1, 0$ if

$$f(x) = \begin{cases} x^2 + x - 5 & \text{if } x \in (-\infty, 0) \\ x^2 + 3x - 2 & \text{if } x \in (3, \infty) \\ x^2 & \text{if } x \in (0, 2) \\ x^2 - 3 & \text{otherwise} \end{cases}$$

4. State whether the following relations are functions or not. If it is a function check for one-to-oneness and ontoness. If it is not a function, state why?
   (i) If $A = \{a, b, c\}$ and $f = \{(a, c), (b, c), (c, b)\}; (f : A \to A)$.
   (ii) If $X = \{x, y, z\}$ and $f = \{(x, y), (x, z), (z, x)\}; (f : X \to X)$.
5. Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d\}$. Give a function from $A \to B$ for each of the following:
   (i) neither one-to-one nor onto. (ii) not one-to-one but onto.
   (iii) one-to-one but not onto. (iv) one-to-one and onto.
6. Find the domain of $\dfrac{1}{1 - 2\sin x}$.
7. Find the largest possible domain of the real valued function $f(x) = \dfrac{\sqrt{4 - x^2}}{\sqrt{x^2 - 9}}$.
8. Find the range of the function $\dfrac{1}{2\cos x - 1}$.
9. Show that the relation $xy = -2$ is a function for a suitable domain. Find the domain and the range of the function.
10. If $f, g : \mathbb{R} \to \mathbb{R}$ are defined by $f(x) = |x| + x$ and $g(x) = |x| - x$, find $g \circ f$ and $f \circ g$.